

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – X

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Class: TYIT	Batch:1
Date of Assignment:03/09/26	Date/Time of Submission:03/09/26

Aim

To perform data preparation and transformation using Tableau Prep Builder by connecting, cleaning, standardizing, merging, aggregating, filtering and exporting student assessment data.

Software Required

- Tableau Prep Builder
- Student_Assessments_Term1.csv
- Student_Assessments_Term2.csv
- Course_Credit_Master.csv

Procedure

A. Connect and Load Data

1. Open **Tableau Prep Builder** and connect the three CSV files using **Connections** → **+** → **Text File**.

Result :

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists three text files: 'Student_Assessment...', 'Student_Assessment...', and 'Course_Credit_Mast...'. The 'Course_Credit_Mast...' file is selected. Below it, the 'Tables' pane shows 'Course_Credit_Master'. The main workspace displays a flow diagram with three data sources connected to a 'View and clean data' step. The 'Input' pane shows 5 fields: 'Source Row File...', 'Subject_Code', 'Subject_Title', 'Credit_Hours', and 'Passing_Marks'. The 'Preview' pane shows a table with 8 rows of data.

#	Source Row File...	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
1					
2		CS101	Data Structures	4	40
3		CS201	Algorithms	4	40
4		EC101	Digital Logic	3	40
5		EC201	Signals & Systems	4	40
6		IT101	Database Systems	4	40
7		IT201	Cloud Computing	4	40
8		ME101	Thermodynamics	3	40

2. Verify **Comma** as the field separator and select **First row contains headers**.

Result :

The screenshot shows the 'Course_Credit_Master' data source settings in Tableau Prep Builder. The 'Header Options' section has 'Set header' selected. The 'Header and Data Start Row' section has 'Header row' set to 1 and 'Data start row' set to 2. The 'Text Options' section has 'Field Separator' set to 'Comma', 'Text Qualifier' set to 'Automatic', 'Character Set' set to 'UTF-8', and 'Locale' set to 'English (United States)'. The 'Preview' pane shows the same table as the previous screenshot.

#	Source Row File...	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
1					
2		CS101	Data Structures	4	40
3		CS201	Algorithms	4	40
4		EC101	Digital Logic	3	40
5		EC201	Signals & Systems	4	40
6		IT101	Database Systems	4	40
7		IT201	Cloud Computing	4	40
8		ME101	Thermodynamics	3	40

B. Clean and Profile Data

1. Add a **Clean Step** to Term 1 and inspect the **Profile Pane**.
2. Apply **Trim Spaces**, standardize student-name casing and change Exam_Date to **Date**.
3. Add a Clean Step to Term 2 and clean Full_Name and Assessment_Date.

Result :

Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
CS101	Data Structures	4	40
CS201	Algorithms	4	40
EC101	Digital Logic	3	40
EC201	Signals & Systems	4	40
IT101	Database Systems	4	40
IT201	Cloud Computing	4	40
ME101	Thermodynamics	3	40

Clean Term 1 :

Student_Assessments_Term1 → student-name → (Trim Spaces)

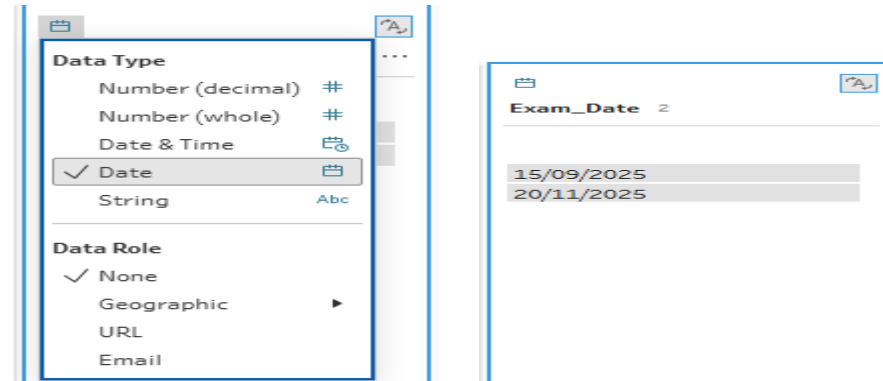
Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Marks_Percentage
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35	87.5
S102	Aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55	83.3
S103	Ananya Singh	CS		2025-09-15	IT101	55	83.3
S104	Kavya Reddy	CSE		2025-11-20	ME101	75	93.8
S105	Priya Verma	ECE		2025/09/15		75	93.8
S106	Rahul Nair	Electronics		2025/09/15		75	93.8
S107	Rohan Gupta	Electronics Engg		2025/11/20		95	106.3
S108	Sneha Patel	Info Tech				95	106.3

Student_Assessments_Term1 → student-name → (Make Uppercase)

Clean 1 9 fields 16 rows

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject	Marks_Obtained
S101	AARAV SHARMA	Comp Sci	Final	15-09-2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20-11-2025	EC101	60
S103	ANANYA SINGH	CS		2025.09.15	IT101	75
S104	KAVYA REDDY	CSE		2025.11.20		
S105	PRIYA VERMA	ECE		2025/09/15	ME101	75
S106	RAHUL NAIR	Electronics		2025/11/20		
S107	ROHAN GUPTA	Electronics Engg				
S108	SNEHA PATEL	Info Tech				
		Information Tech				
		IT				
		Mech Engg				

Student_Assessments_Term1 → Exam_Date → Change Data Type Date

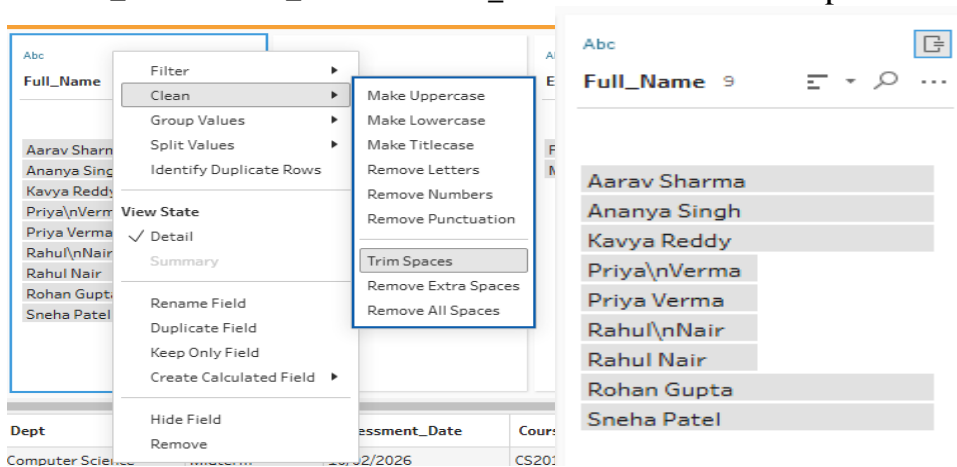


Clean Term 2 :

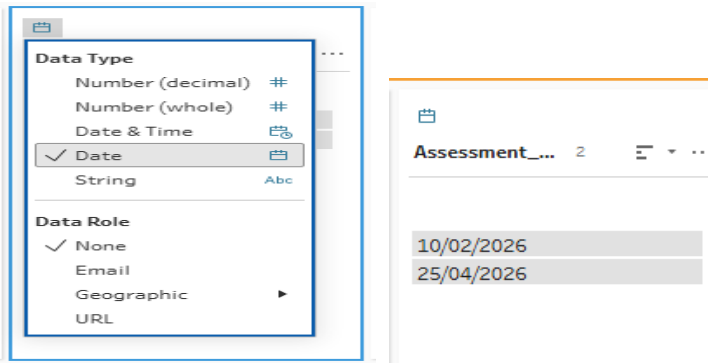
Clean 2 9 fields 14 rows

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201	98	100	Active
S101	Ananya Singh	CS	Midterm	25/04/2026	CS201	95	100	Active
S102	Kavya Reddy	CSE	Midterm	10/02/2026	EC201	75	100	Active
S103	PriyaVerma	ECE			IT201	88		
S105	Priya Verma	Electronics				94		
S107	Rahul Nair	Information Tech				96		
S108	Rohan Gupta	IT				98		
	Sneha Patel					95		

Student_Assessments_Term2 → Full_Name → Clean → Trim Spaces.



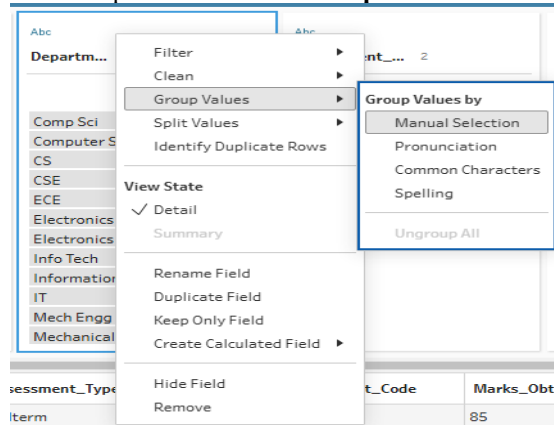
Student_Assessments_Term2 → Assessment_Date. → Change Data Type → Date



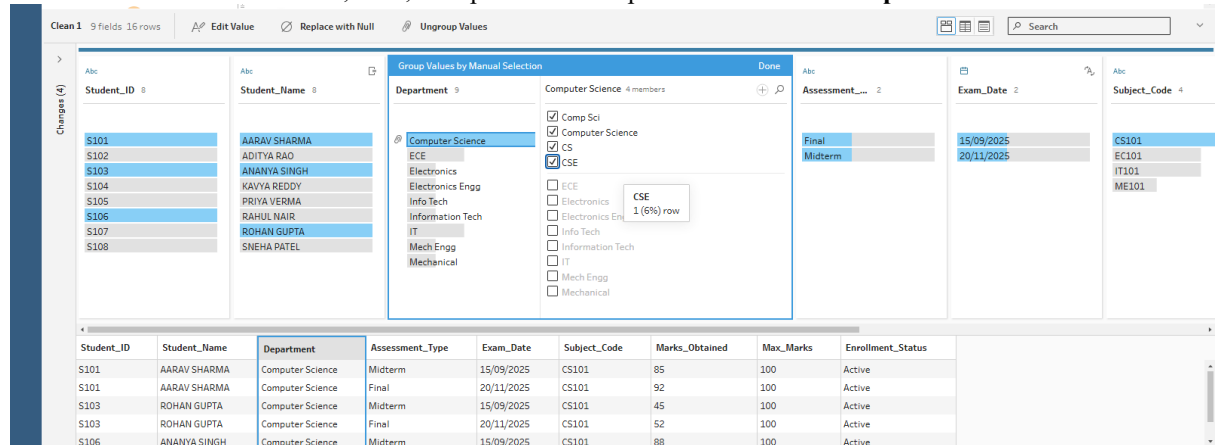
C. Group and Standardize Values

Clean 1 :

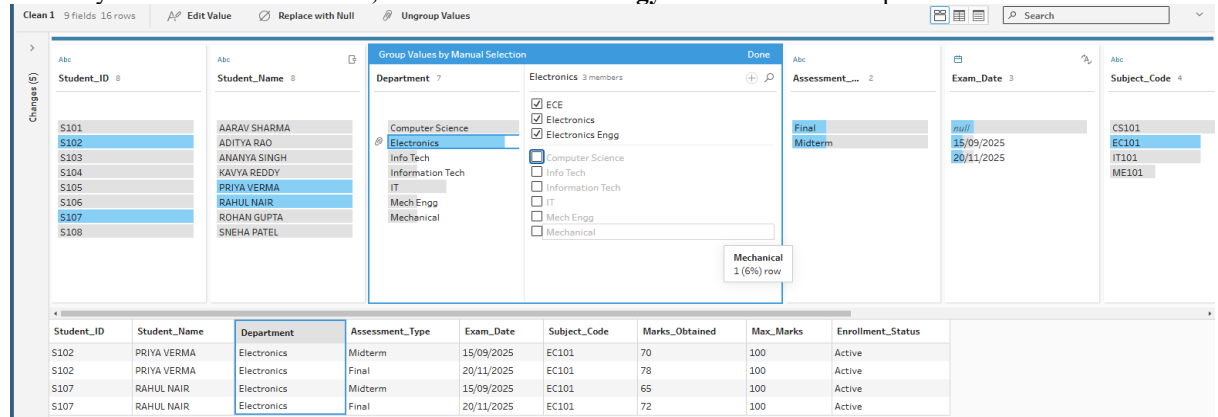
1. Select Department and use **Group Values**.



2. Combine variations such as CS, CSE, Comp Sci and Computer Science into **Computer Science**.



3. Similarly standardize **Electronics, Information Technology and Mechanical** departments.



4. Combine variations such as **IT, Information Tech, and Info Tech** into **Information Technology (IT)**.

Clean 1 9 fields 16 rows

Group Values by Manual Selection

Department 5

Information Tech 3 members

Computer Science
Electronics
Information Tech
Mech Engg
Mechanical

Info Tech
Information Tech
IT

Computer Science
Electronics
Mech Engg
Mechanical

Student_ID 8

Student_Name 8

Assessment_Type 2

Exam_Date 2

Subject_Code 4

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Information Tech	Midterm	15/09/2025	IT101	90	100	Active
S102	ADITYA RAO	Information Tech	Final	20/11/2025	IT101	96	100	Active
S103	ANANYA SINGH	Information Tech	Midterm	15/09/2025	IT101	92	100	Active
S104	KAVYA REDDY	Information Tech	Final	20/11/2025	IT101	98	100	Active
S105	PRIYA VERMA							
S106	RAHUL NAIR							
S107	ROHAN GUPTA							
S108	SNEHA PATEL							

5. Combine variations such as **Mechanical and Mech Engg** into **Mechanical (ME)**.

Clean 1 9 fields 16 rows

Group Values by Manual Selection

Depta... 4

Mechanical 2 members

Computer Science
Electronics
Information Tech
Mechanical

Mech Engg
Mechanical

Mechanical 2 (13%) rows
2 (100%) highlighted

Student_ID 8

Student_Name 8

Assessment_Type 2

Exam_Date 3

Subject_Code 4

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S105	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
S105	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped

Clean 2 :

1. Combine variations such as **CS, CSE, Comp Sci, and Computer Science** into **Computer Science (CS)**.

Clean 2 9 fields 14 rows

Group Values by Manual Selection

Dept 5

Computer Science 3 members

Computer Science
CS
CSE

ECE
Electronics
Information Tech
IT

ECE
Electronics
Information Tech
IT

Std_ID 7

Full_Name 9

Exam_Type 2

Assessment_Date 2

Course_Code 3

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S106	Ananya Singh	Computer Science	Final	25/04/2026	CS201	96	100	Active
S101	Aarav Sharma	Computer Science	Midterm	10/02/2026	CS201	88	100	Active
S103	Rohan Gupta	Computer Science	Final	25/04/2026	CS201	60	100	Active
S101	Aarav Sharma	Computer Science	Final	25/04/2026	CS201	95	100	Active

2. Combine variations such as **ECE, Electronics, and Electronics Engg** into **Electronics (EC)**.

Clean 2 9 fields 14 rows

Group Values by Manual Selection

Dept 4

Electronics 2 members

Computer Science
Electronics
Information Tech
IT

ECE
Electronics

Computer Science
Information Tech
IT

Std_ID 7

Full_Name 9

Exam_Type 2

Assessment_Date 2

Course_Code 3

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S107	Rahul/Nair	Electronics	Final	25/04/2026	EC201	75	100	Active
S102	Priya/Verma	Electronics	Final	25/04/2026	EC201	82	100	Active
S107	Rahul Nair	Electronics	Midterm	10/02/2026	EC201	70	100	Active
S102	Priya Verma	Electronics	Midterm	10/02/2026	EC201	75	100	Active

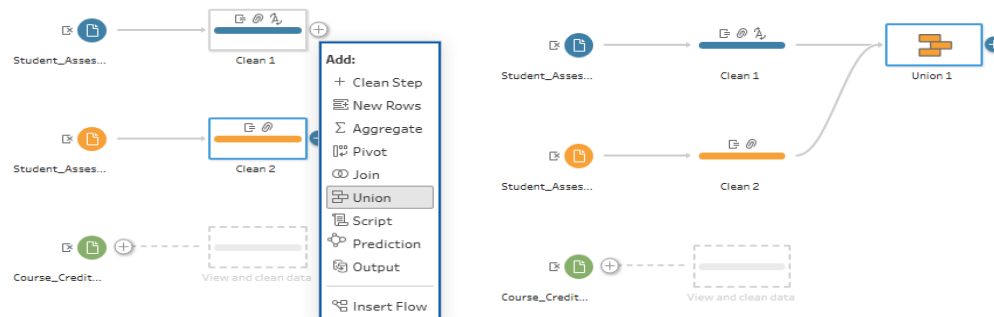
3. Combine variations such as **IT, Information Tech, and Info Tech** into **Information Technology (IT)**.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S104	Sneha Patel	Information Tech	Final	25/04/2026	IT201	98	100	Active
S108	Kavya Reddy	Information Tech	Midterm	10/02/2026	IT201	95	100	Active
S108	Kavya Reddy	Information Tech	Final	25/04/2026	IT201	100	100	Active
S104	Sneha Patel	Information Tech	Midterm	10/02/2026	IT201	94	100	Active

D. Merge Mismatched Fields

1. Create a **Union** between the cleaned Term 1 and Term 2 datasets.
2. Match the mismatched fields:
 - Std_ID → Student_ID
 - Full_Name → Student_Name
 - Dept → Department
 - Exam_Type → Assessment_Type
 - Assessment_Date → Exam_Date
 - Course_Code → Subject_Code
 - Score → Marks_Obtained
 - Total_Possible → Max_Marks

Result :



Union 1 17 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

14 Mismatching fields from 17 resulting fields.

Mismatched Fields

- Full_Name
- Student_Name
- Dept
- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible

Union Results Show only mismatched fields

Drop to merge fields

Student_Name Department Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Full_Name Dept

Union 1 16 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

12 Mismatching fields from 16 resulting fields.

Mismatched Fields

- Dept
- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Department
- Assessment_Type

Union Results Show only mismatched fields

Drop to merge fields

Department Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Dept Exam_Type Assessment_Date

Union 1 15 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

10 Mismatching fields from 15 resulting fields.

Mismatched Fields

- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Assessment_Type
- Exam_Date
- Subject_Code

Union Results Show only mismatched fields

Drop to merge fields

Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Exam_Type Assessment_Date Course_Code Score Total_Possible

Union 1 14 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Rename Field Create Calculated Field... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

8 Mismatching fields from 14 resulting fields.

Mismatched Fields

- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Exam_Date
- Subject_Code
- Marks_Obtained
- Max_Marks

Union Results Show only mismatched fields

Drop to merge fields

Exam_Date Subject_Code Marks_Obtained Max_Marks Assessment_Date Course_Code Score Total_Possible

Union 1 13 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

6 Mismatching fields from 13 resulting fields.

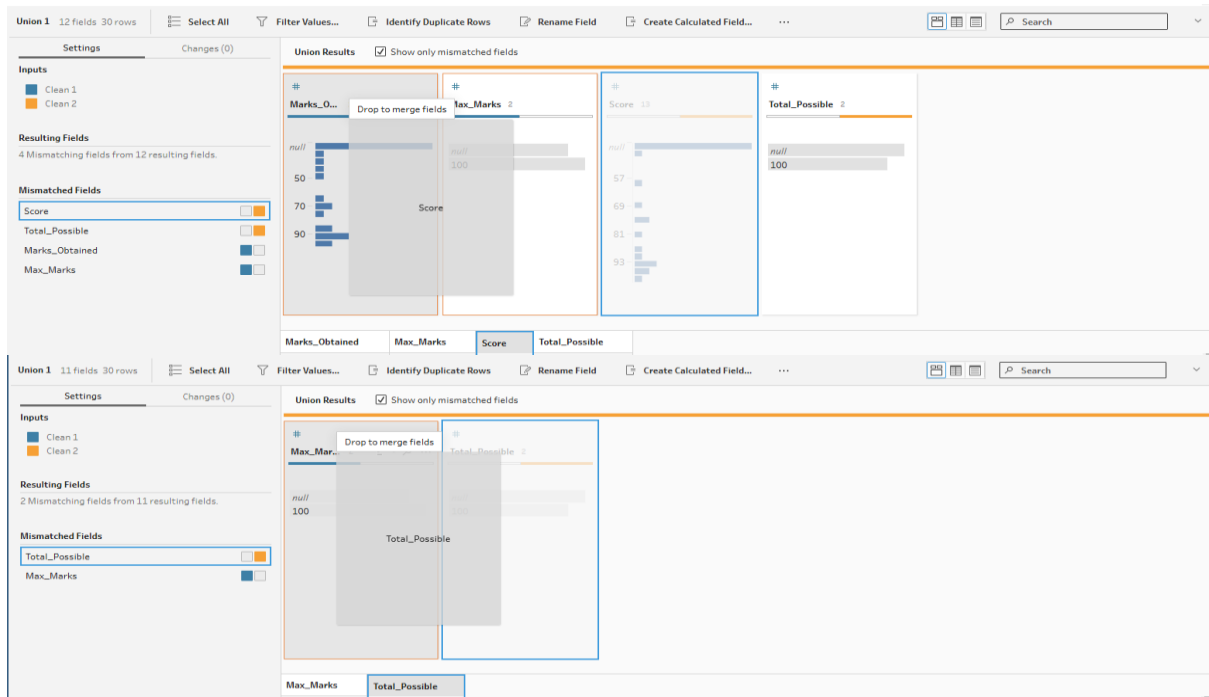
Mismatched Fields

- Course_Code
- Score
- Total_Possible
- Subject_Code
- Marks_Obtained
- Max_Marks

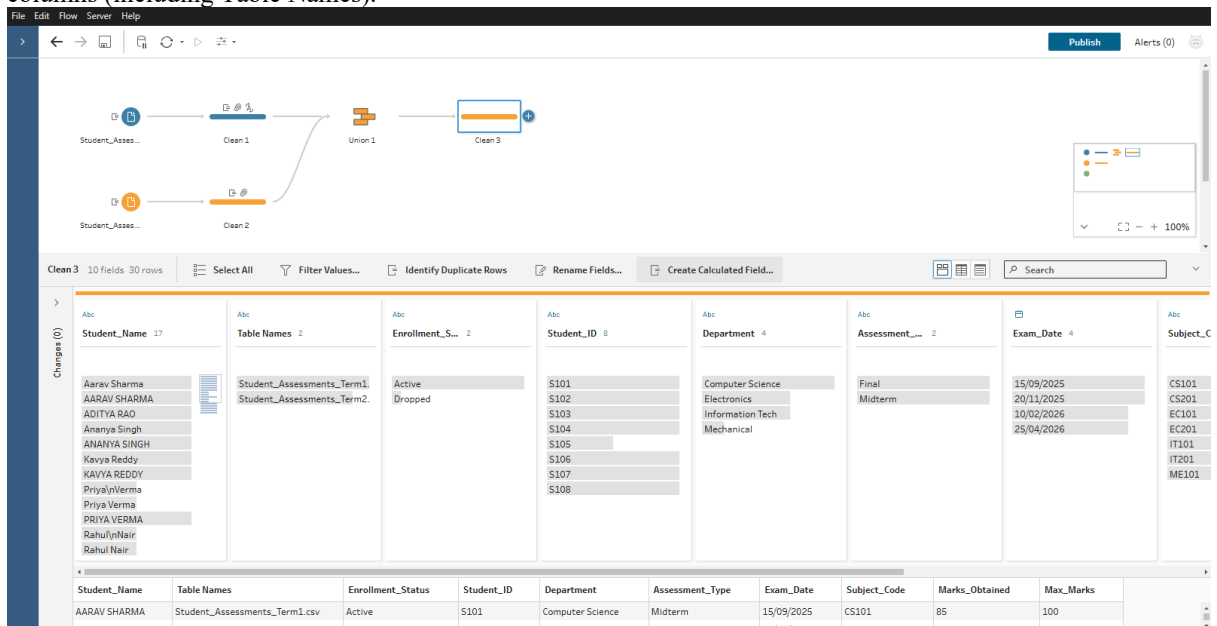
Union Results Show only mismatched fields

Drop to merge fields

Subject_Code Marks_Obtained Max_Marks Course_Code Score Total_Possible



Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).



E. Perform Aggregation

1. Add an **Aggregate Step** after the Union.
2. Group the data by **Student_ID**.
3. Calculate **Total Marks, Average Marks and Assessment Count**.

Result :

Field Name

Term

IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END

All

Search

ABS
ACOS
AND
AREA
ASC
ASCII
ASIN
ATAN
ATAN2
AVG
BUFFER
CASE
CEILING
CHAR
CONTAINS
COS
COT
COUNT
COUNTD

ABS(number)

Returns the absolute value of the given number.
Example: ABS(-7) = 7

Calculation is valid

Create with Tableau Agent

Apply

Apply and Close

File Edit View Server Help

← → ↺ ↻ 🔍

Publish Alerts (0)

Student_Asses... Clean 1 Union 1 Clean 3

Student_Asses... Clean 2

Add:
+ Clean Step
New Rows
Aggregate
Pivot
Join
Union
Script
Prediction
Output
Insert Flow

Clean 3 11 fields 30 rows

Select All Filter Values... Identify Duplicate Rows

Split Custom Split... Rename Field

Search

Changes (1)

Term 2

Student_Name 17

Table Names 2

Enrollment_Status 2

Student_ID 8

Department 4

Assessment_Type 2

Exam_Date

Term 1 AARAV SHARMA Student_Assessments_Term1.csv Active S101 Computer Science Midterm 15/09/2025 CS101 85 100

Term 1 AADAV SHARMA Student_Assessments_Term2.csv Active S101 Computer Science Midterm 20/11/2025 CS101 93 100

Aggregate 1 7 fields 17 rows

Select All Filter Values... Identify Duplicate Rows Keep Only Field Remove Field

Settings Changes (0)

Additional Fields

Drag fields to aggregate or group them.

Search

Add All Remove All

GROUP Enrollment_Status
GROUP Exam_Date
GROUP Subject_Code
GROUP Table Names
SUM Number of Rows (Aggregated)

Grouped Fields

GROUP GROUP GROUP

Student_Name 17 Student_ID 8

Aarav Sharma S101
AARAV SHARMA S102
ADITYA RAO S103
Ananya Singh S104
ANANYA SINGH S105
Kavya Reddy S106
KAVYA REDDY S107
Priya Verma S108
PRIYA VERMA
Rahul Nair

Aggregated Fields

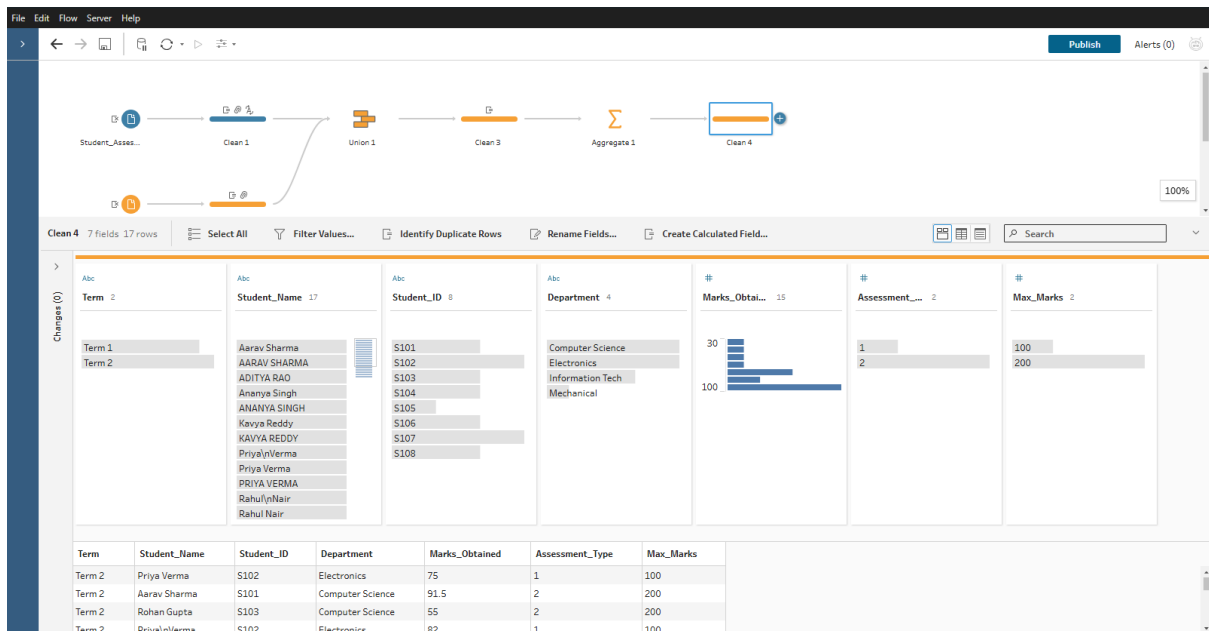
AVG # CNT # SUM

Marks_Obtain... 15 Assessment_Type 2 Max_Marks 2

75 1 100
91.5 2 200

Term Student_Name Student_ID Department

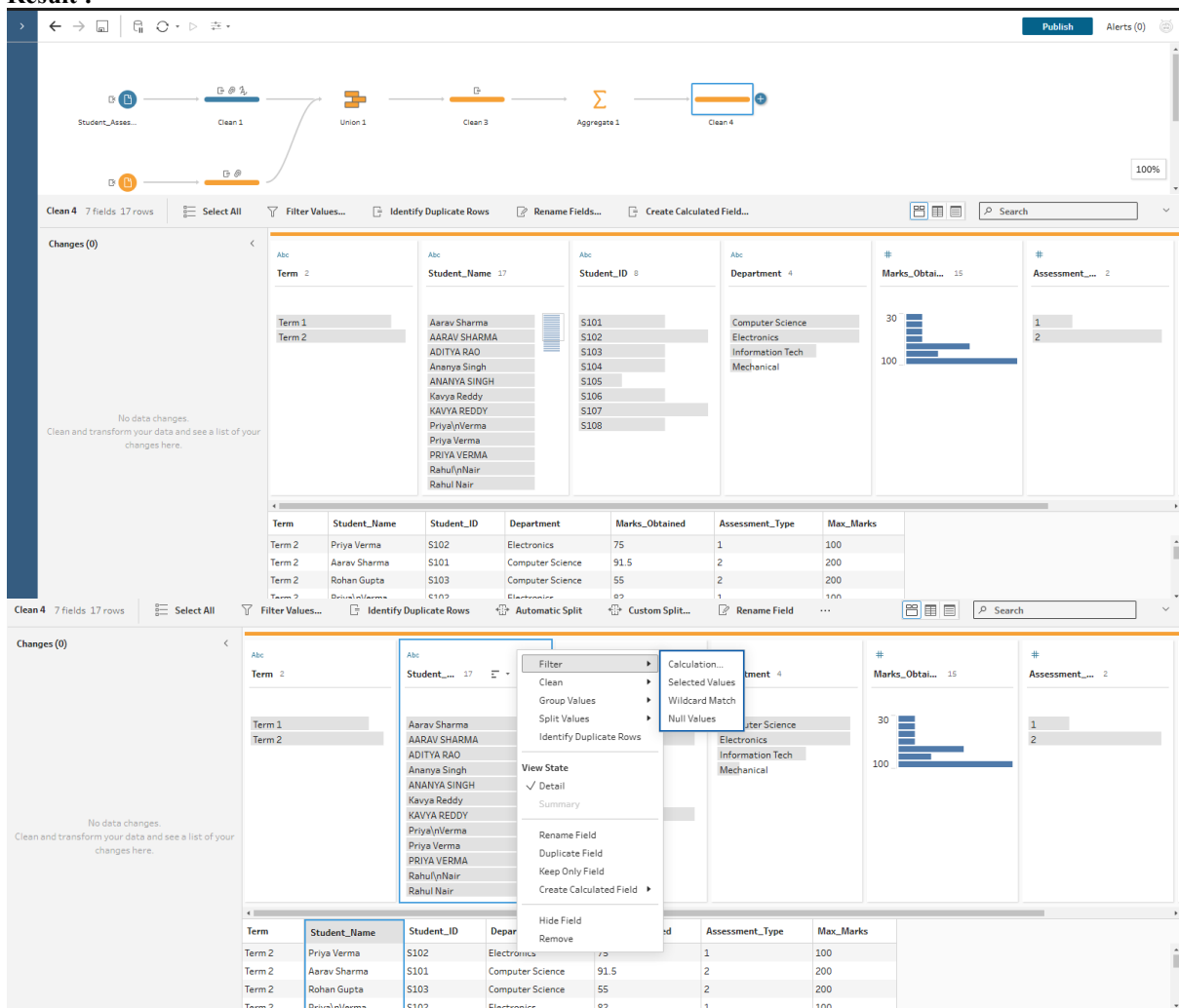
Term 2 Priya Verma S102 Electronics
Term 2 Aarav Sharma S101 Computer Science



F. Apply Filters and Transformations

1. Filter Enrollment_Status and keep only **Active** students.
2. Create a Percentage calculated field using Marks Obtained and Maximum Marks.
3. Create Grade and Result_Status calculated fields based on Percentage.

Result :



The screenshot shows two 'Add Field' dialog boxes in Tableau. The first dialog box has 'Letter_Grade' as the field name and contains a script that maps percentage ranges to letter grades: A (Distinction) for >= 85, B (First Class) for >= 75, C (Second Class) for >= 60, D (Pass Class) for >= 40, and F (Fail / Re-appear) for < 40. The second dialog box has 'Academic_Standing' as the field name and contains a script that maps percentage ranges to academic standing: Satisfactory for >= 60 and Academic Probation (<60%) for < 60. Below the dialog boxes is a data preview table with 10 rows and 4 columns: Letter_Grade, Term_Percentage, Academic_Standing, and Total_Marks.

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks
A (Distinction)	42	Academic Probation (<60%)	42
A (Distinction)	45	Satisfactory	45
C (Second Class)	71	Satisfactory	88
C (Second Class)	74	Satisfactory	95
C (Second Class)	88	Satisfactory	142
C (Second Class)	88.5	Satisfactory	148
C (Second Class)	91	Satisfactory	177
C (Second Class)	95	Satisfactory	182

G. Join and Output

1. Join the prepared assessment data with Course_Credit_Master.csv using Subject_Code.
2. Add an **Output Step** and select CSV or Tableau Hyper Extract.
3. Run the flow and verify the output.

Result :

The screenshot shows the 'Add' menu in Tableau Desktop. The 'Output' option is selected, which will create an output step in the data flow. The background shows a data preview table with columns: Term, Full_Name, and Subject_Code.

Term	Full_Name	Subject_Code
Term 1	Full Name 1	Subject Code 1
Term 2	Full Name 2	Subject Code 2
Term 3	Full Name 3	Subject Code 3
Term 4	Full Name 4	Subject Code 4
Term 5	Full Name 5	Subject Code 5

Output 10 fields

Save output to: File

Name: Student_Performance_Master_Cleaned

Location: C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Datasources

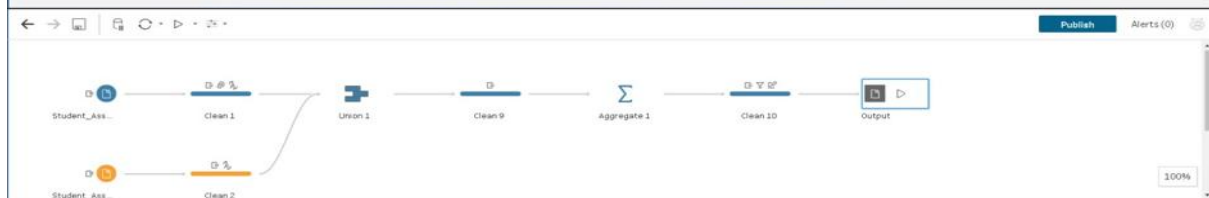
Output type: Tableau Data Extract (.hyper)

Write Options
Select an option to create or update your output table.
Full refresh
Create table

Run Flow

Save to Output.hyper

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks_Obtained	Term	Full_Name	Student_ID	Dept	Assessm
D (Pass Class)	45	Academic Probation (<60%)	45	Term 2	Aditya Rao	S105	Mechanical	1
C (Second Class)	74	Satisfactory	148	Term 1	PRIYA VERMA	S102	Electronics	2
A (Distinction)	88	Satisfactory	88	Term 2	Aarav Sharma	S101	CS	1
C (Second Class)	71	Satisfactory	142	Term 1	NEHA JOSHI	S108	Information Tech	2
D (Pass Class)	42	Academic Probation (<60%)	42	Term 2	Aditya Rao	S105	Mech Engg	1
A (Distinction)	91	Satisfactory	182	Term 1	ANANYA SINGH	S106	Computer Science	2
A (Distinction)	88.5	Satisfactory	177	Term 1	AARAV SHARMA	S101	Computer Science	2
A (Distinction)	95	Satisfactory	95	Term 2	Aarav Sharma	S101	Computer Science	1



Output 10 fields

Save output to: File

Name: Student_Performance_Master_Cleaned

Location: C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Datasources

Output type: Comma Separated Values (.csv)

Write Options
Select an option to create or update your output table.
Full refresh
Create table

Run Flow

Save to Student_Performance_Master_Cleaned.csv

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks_Obtained	Term	Full_Name	Student_ID	Dept	Assessm
D (Pass Class)	45	Academic Probation (<60%)	45	Term 2	Aditya Rao	S105	Mechanical	1
C (Second Class)	74	Satisfactory	148	Term 1	PRIYA VERMA	S102	Electronics	2
A (Distinction)	88	Satisfactory	88	Term 2	Aarav Sharma	S101	CS	1
C (Second Class)	71	Satisfactory	142	Term 1	NEHA JOSHI	S108	Information Tech	2
D (Pass Class)	42	Academic Probation (<60%)	42	Term 2	Aditya Rao	S105	Mech Engg	1
A (Distinction)	91	Satisfactory	182	Term 1	ANANYA SINGH	S106	Computer Science	2
A (Distinction)	88.5	Satisfactory	177	Term 1	AARAV SHARMA	S101	Computer Science	2
A (Distinction)	95	Satisfactory	95	Term 2	Aarav Sharma	S101	Computer Science	1

Finished Running Flow

Student_Performance_Master_Cleaned.csv

Total time 00:02

Done

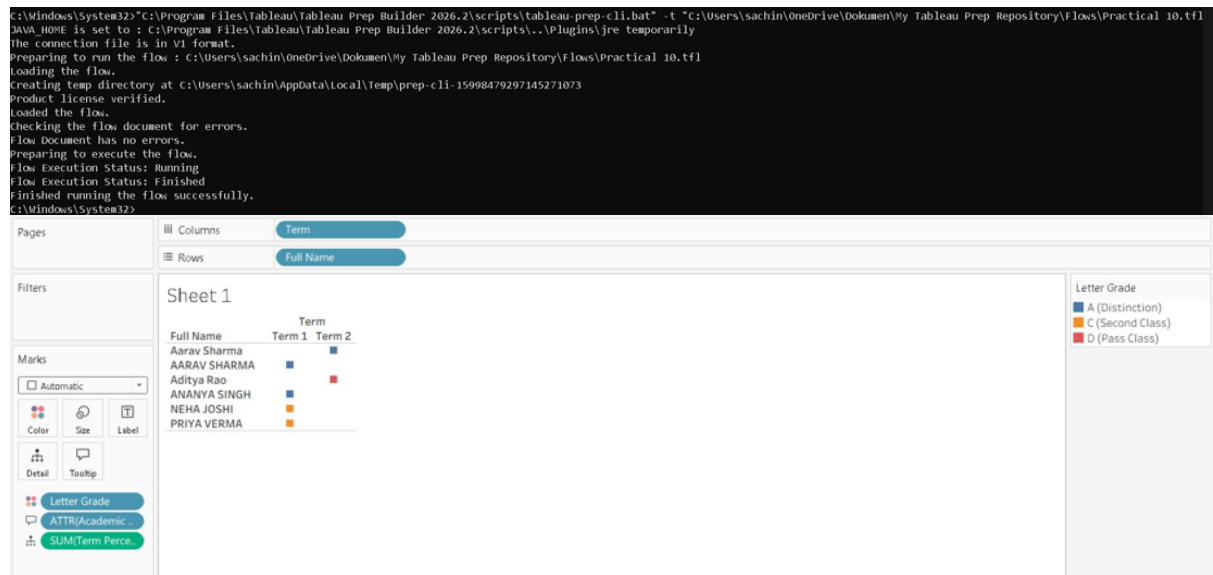
Term	Full_Name	Student_ID	Dept	Assessm
Term 2	Aditya Rao	S105	Mechanical	1
Term 1	PRIYA VERMA	S102	Electronics	2
Term 2	Aarav Sharma	S101	CS	1
Term 1	NEHA JOSHI	S108	Information Tech	2
Term 2	Aditya Rao	S105	Mech Engg	1
Term 1	ANANYA SINGH	S106	Computer Science	2
Term 1	AARAV SHARMA	S101	Computer Science	2
Term 2	Aarav Sharma	S101	Computer Science	1

```
C:\Windows\System32>dir "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\*.tfl" /s
Volume in drive C is Windows-SSD
Volume Serial Number is D4E0-1C17
```

```
Directory of C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows
```

```
04-09-2026 12:43      28,734 Practical 10.tfl
04-09-2026 12:43      28,734 Student_Performance_Prep_Flow.tfl
                2 File(s)          57,468 bytes
```

```
Total Files Listed:
                2 File(s)          57,468 bytes
                0 Dir(s) 189,394,382,848 bytes free
```



Result

Successfully created a Tableau Prep data preparation flow to connect, clean, standardize, merge, aggregate, filter and transform student assessment data, and exported the final prepared dataset.